

NLA Midterm (November 2025) Variant A

Problem 1. (1 + 2 pts)

- (a) Find a rotation matrix A of the (x, z) plane in $\mathbb{R}^3$ with $A^7 = I$.
- (b) Find the eigendecomposition of A .

Problem 2. (3 + 1 pts)

$$A = \begin{bmatrix} 25 & 50 & -74 \\ 4 & -8 & 13 \\ 6 & 12 & -17 \end{bmatrix}$$

- (a) Compute A^{2025} .
- (b) Find the spectral decomposition of A , i.e. $A = \sum_{\lambda \in \text{spec}(A)} \lambda P_\lambda$, where $\text{spec}(A)$ is the set of (distinct) eigenvalues of A , and P_λ is the orthogonal projector onto the eigenspace corresponding to λ .

Problem 3. (2 + 1 pts)

$$A = \begin{bmatrix} 2 & 5 & 6 \\ 4 & 10 & 13 \\ 4 & 11 & 13 \end{bmatrix}$$

- (a) Find the LU decomposition of A (with pivoting if necessary).
- (b) Find the Skeleton decomposition of A .

Problem 4. (1 + 3 pts)

You are given the three data points: $(-1, 14), (1, 14), (2, 7)$.

- (a) Formulate the least squares problem to find the best-fit line.
- (b) Find the minimum ℓ_2 -norm least squares solution.

Problem 5. (2 pts)

Let $A \in \mathbb{C}^{n \times n}$ and let $M, L \subset \mathbb{C}^n$ be m -dimensional subspaces such that A is non-singular and $L = AM$. Let $V, W \in \mathbb{C}^{n \times m}$ be bases for M and L , respectively. Show that the matrix $W^*AV \in \mathbb{C}^{m \times m}$ is non-singular.

Problem 6. (3 + 1 pts)

- (a) Let $A \in \mathbb{C}^{n \times n}$, $b \in \mathbb{C}^n$, $x_0 \in \mathbb{C}^n$ and let $M \subset \mathbb{C}^n$ be a subspace. Show that a vector x satisfies

$$x \in x_0 + M, \quad b - Ax \perp AM$$

if and only if

$$x = \arg \min_{y \in x_0 + M} \|b - Ay\|_2.$$

- (b) Show that

$$b - Ax = (I - P)(b - Ax_0),$$

where P is the orthogonal projector onto the subspace AM .